Extrinsic Calibration of LiDAR and Fisheye Camera Based on Deep Learning and Geometric Optimization

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Abstract: Extrinsic calibration plays a crucial role in autonomous driving, intelligent surveillance, and robotic perception. However, existing research primarily focuses on the calibration between LiDAR and pinhole cameras, while studies on LiDAR and fisheye cameras remain relatively limited. To fill this gap, this paper proposes a deep learning-based extrinsic calibration method for a LiDAR and a fisheye camera, incorporating a geometric optimization strategy as post-processing to significantly improve calibration accuracy. The proposed method first utilizes a neural network to predict the extrinsic transformation matrix, followed by geometric optimization to refine the predicted results. Experimental evaluations on 18,335 image–point cloud pairs demonstrate that the deep learning model reduces the point cloud alignment error by 91.6%. With the additional geometric post-processing, the error is further reduced by 52.4%, verifying the high accuracy, reliability, and robustness of the proposed approach in practical applications.

Key words: Extrinsic calibration; deep learning; fisheye camera; geometric optimization

In recent years, autonomous driving technology has made significant progress, with environmental perception as the core module directly determining the safety and reliability of the system [1]. Multi-sensor fusion has become a key means to enhance perception capabilities [2], particularly the combination of LiDAR and cameras, which not only provides high-precision depth information but also enhances scene understanding capabilities. However, since various sensors have their own independent coordinate systems, achieving effective data fusion depends on high-precision extrinsic calibration.

Traditional calibration methods mostly adopt artificial auxiliary tools such as checkerboards and calibration frames [3], typically applied in static environments and mostly completed before vehicle delivery. According to whether calibration boards are used, they can be divided into target-based and targetless methods. The former is represented by checkerboards, for example, Zhengyou Zhang [4] constructed a classic camera intrinsic calibration framework by collecting corner images from multiple angles to solve parameters. Based on this, Qilong Zhang et al. [5] introduced nonlinear optimization to further improve calibration accuracy. Huaiyu Cai et al. [6] designed a new calibration board, combining local gradient information with main plane corner information, significantly improving the calibration accuracy between LiDAR and cameras.

Targetless calibration methods mainly rely on sensor motion trajectories or environmental natural features to estimate extrinsic parameters. For example, Klaus H. Strobl and Gerd Hirzinger [7] proposed a method for solving extrinsic parameters between cameras and robotic arms based on preset motion trajectories of robotic arms, widely applied in robotic systems. Chongjian Yuan et al. [8] proposed an extrinsic estimation method based on natural scene edge feature alignment, achieving sub-pixel accuracy.

Although static calibration methods have good effects in the initial deployment stage, in practical applications, factors such as mechanical vibration, temperature changes, and minor collisions may cause sensor installation attitude drift, leading to extrinsic misalignment and affecting system perception performance. Therefore, online extrinsic calibration methods with adaptive capabilities have gradually become research hotspots. In recent years, the widespread application of deep learning in object detection [9], trajectory prediction [10], and other fields has provided new ideas for extrinsic calibration, making online calibration more robust and real-time in dynamic scenes.

Online calibration methods are mainly divided into optimization-based and deep learning-based categories.

The former relies on geometric or semantic features in scenes to establish objective functions for minimization solving. For example, Yufeng Zhu et al. [11] proposed a semantic calibration method based on mutual information optimization; Hao Wu et al. [12] integrated lane line and vehicle information to enhance the stability of extrinsic parameter calculation. The latter uses neural networks to learn spatial mapping relationships between images and point clouds, directly predicting extrinsic parameters and avoiding complex feature matching and optimization processes. For example, RegNet proposed by Nick Schneider et al. [13] integrates feature extraction, matching, and extrinsic regression into an end-to-end network, achieving real-time calibration; CalibNet proposed by Ganesh Iyer et al. [14] also has good real-time performance. Subsequent models such as RGGNet [15], CalibRCNN [16], and LCCNet [17] further improved the accuracy and stability of extrinsic estimation.

Currently, most research still focuses on calibration between LiDAR and pinhole cameras, while research on extrinsic calibration between LiDAR and fisheye cameras is relatively scarce. Jack Borer et al. [18] proposed using monocular depth estimation and LiDAR geometric constraints for optimization, but this method may have significant errors in the depth estimation stage, affecting overall accuracy.

Considering that fisheye cameras have ultra-wide viewing angles and can effectively cover near-distance blind spots, they are particularly suitable for surrounding perception of autonomous vehicles in complex urban environments. To intuitively demonstrate the advantages of fisheye cameras, as shown in Figure 1, we compared the imaging effects of the same physical camera (sharing the same intrinsic parameters) when using fisheye models and pinhole models for a person located at the same position. The results show that the fisheye camera model can significantly provide a wider field of view, especially in the edge areas of the image, clearly capturing the full-body image of the person, while the pinhole camera model can only capture partial areas. This fully demonstrates that fisheye cameras can provide a wider field of view, reduce visual blind spots, and have obvious advantages for scenes requiring wide-range perception.

Fig.1 Comparison of Imaging Effects between Fisheye Camera and Traditional Pinhole Camera

This paper focuses on the extrinsic calibration problem between LiDAR and fisheye cameras in automatic following vehicles, with the experimental platform shown in Figure 2. The platform mounts fisheye cameras and LiDAR at the front of the vehicle, aiming to achieve automatic following of cleaning personnel to assist in their cleaning operations. In the method design of this paper, we use synchronized image and point cloud information for neural network training and further optimize prediction results through geometric optimization post-processing strategies. This method can achieve high-precision extrinsic estimation in dynamic environments, thereby enhancing the environmental perception capability and driving safety of autonomous driving systems.

Fig.2 Schematic Diagram of the Experimental Platform

1 Sensor Extrinsic Calibration Method Design

1.1 Data Acquisition and Preprocessing

1) Training Sample Construction:

Since directly calibrating extrinsic parameters for each pair of synchronized images and point clouds frame by frame is extremely challenging, this paper constructs a large number of training samples by transforming each pair of synchronized image and point cloud data. The specific process is as follows:

For each synchronized data pair $pair_l = \{I, P_l\}$, where I represents the fisheye camera image and, P_l represents point cloud data in the LiDAR coordinate system, we first use the known initial extrinsic matrix M_{init} to convert P_l to the fisheye camera coordinate system, obtaining the converted point cloud P_f:

$$P_f^{(h)} = M_{init} \cdot P_l^{(h)} \tag{1}$$

where $P_f^{(h)}$ and $P_l^{(h)}$ represent the homogeneous coordinate forms of point cloud data in the LiDAR and fisheye camera coordinate systems, respectively; $[\cdot]^{(h)}$ represents the homogeneous coordinate expression form.

Based on this, training data is generated by applying new extrinsic transformations M_{label}. The specific transformation process is shown in equations (2) and (3):

$$P_{train}^{(h)} = M_{label} \cdot P_f^{(h)} \tag{2}$$

$$P_{train} = [P_{train}^{(h)}]_{1:3} \tag{3}$$

where M_{label} is the new extrinsic parameter used to generate training data, and its construction method will be elaborated below; P_{train} is the training point cloud data obtained by transforming P_f through M_{label}.

By continuously adjusting the values of M_{label}, this paper can construct a large number of training sample pairs, namely $pair_{train} = \{I, P_{train}\}$, providing sufficient data support for subsequent deep learning model training.

Next, we introduce how to construct M_{label}. This matrix can be represented by six parameters, three of which represent the rotation vector $rvec = [a, b, c]$, and the other three represent the translation vector $tvec = [x, y, z]$. For more intuitive geometric meaning, the rotation vector in this paper adopts spherical coordinate representation, with the formula as follows:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha \begin{bmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{bmatrix} \tag{4}$$

Therefore, M_{label} can be uniquely determined by the six-dimensional parameter vector $m_{label} = [\theta, \varphi, \alpha, x, y, z]$. The value ranges of each parameter are as follows:

$$\begin{gathered} 0 \leq \theta \leq 180° \\ 0 \leq \varphi < 360° \\ -10° \leq \alpha \leq 10° \\ -1 \leq x \leq 1 \\ -1 \leq y \leq 1 \\ -1 \leq z \leq 1 \end{gathered} \tag{5}$$

2) Image Data Preprocessing:

This paper uses original fisheye images with a resolution of 1080 × 1920 × 3 (height × width × channels). Due to significant distortion and edge black borders in the images, preprocessing is required to extract effective information regions. Specific steps include: constructing region of interest masks, cropping effective regions, and adjusting image size to 512 × 512 × 3, thus generating training images I_{train}, as shown in Figure 3.

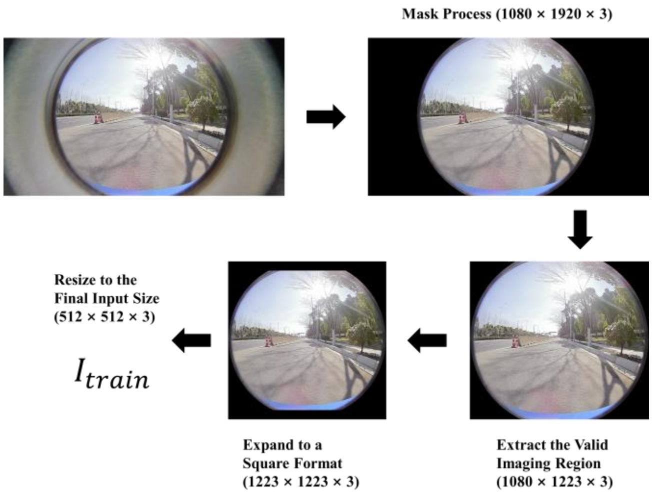

Fig.3 Preprocessing Workflow of Fisheye Images

3) Point Cloud Data Preprocessing: Geometric Mapping Image Construction

To effectively input point cloud information into deep learning networks, this paper adopts the Geometric Mapping Image (GMI) representation method, projecting three-dimensional point clouds into two-dimensional three-channel images.

Let the training point cloud data be $P_{train} = \{ (X_i, Y_i, Z_i) \mid i = 1,2,3, \ldots, N\}$, Based on the Kannala-Brandt fisheye camera projection model [19], the steps for constructing geometric mapping images are as follows:

$$a_i = \frac{X_i}{Z_i}, b_i = \frac{Y_i}{Z_i} \tag{6}$$

$$r_i = \sqrt{a_i^2 + b_i^2} \tag{7}$$

$$\theta_i = \arctan(r_i) \tag{8}$$

$$\theta_{d_i} = \theta_i \cdot (1 + k_1\theta_i^2 + k_2\theta_i^4 + k_3\theta_i^6 + k_4\theta_i^8) \tag{9}$$

$$\tilde{X}_i = \frac{\theta_{d_i}}{r_i} \cdot a_i, \tilde{Y}_i = \frac{\theta_{d_i}}{r_i} \cdot b_i \tag{10}$$

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \tilde{X}_i \\ \tilde{Y}_i \\ 1 \end{bmatrix} \tag{11}$$

subsequently, the projected pixel coordinates (v_i, u_i) are associated with point cloud data P_{train}, to construct GMI, where the three channels are assigned normalized coordinates of the point cloud:

$$GMI(v_i, u_i) = \left(\frac{X_i}{X_{max}}, \frac{Y_i}{Y_{max}}, \frac{Z_i}{Z_{max}}\right) \tag{12}$$

where k_1、k_2、k_3、k_4 are distortion coefficients of the fisheye camera; f_x、f_y、c_x、c_y are camera intrinsic parameters; X_{max}、Y_{max}、Z_{max} are normalization coefficients used to limit the point cloud projection region while ensuring normalization of GMI channel values.

Finally, GMI is generated with a resolution of 1080 × 1920 × 3. To ensure that GMI and I_{train} have the same resolution (512 × 512 × 3), operations similar to the previous image data preprocessing (effective region extraction, square cropping, and scale adjustment) are needed. However, directly scaling the original GMI may lead to point cloud information loss. Therefore, this paper adopts a camera intrinsic matrix transformation method to directly obtain target-sized GMI_{train} from P_{train}.

The transformation process is defined as follows:

$$\begin{bmatrix} f_x' & 0 & c_x' \\ 0 & f_y' & c_y' \\ 0 & 0 & 1 \end{bmatrix} = s \begin{bmatrix} f_x & 0 & c_x - x_{offset} \\ 0 & f_y & c_y - y_{offset} \\ 0 & 0 & \frac{1}{s} \end{bmatrix} \tag{13}$$

where x_{offset}、y_{offset} are translation amounts along x and y axes respectively; s is the scaling factor. Using this transformed intrinsic matrix and executing formulas (6)-(11), GMI_{train} with size 512 × 512 × 3 can be directly generated, maximally preserving point cloud information. Figure 4 shows the differences between two GMI_{train} generation methods: (a) obtained by scaling the original GMI, while (b) is directly generated using the transformed camera intrinsics. From the locally magnified regions, it can be clearly seen that method (a) results in point cloud images with fractures and sparsity, while method (b) has more continuous and dense point cloud distribution, significantly improving information fidelity.

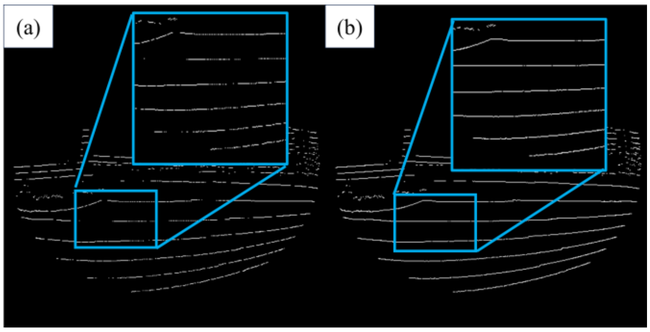

Fig.4 Comparison of Results from Two Different GMI_{train} Generation Methods

1.2 Deep Learning Network

1) Network Structure:

Dual-branch Feature Extraction: This paper constructs a dual-branch deep neural network architecture for processing modal information from fisheye images and point cloud data respectively. The image branch takes training images I_{train} as input, using a pre-trained ResNet18 network as a feature extractor to capture key visual features such as textures, edges, and structures in images [20]. The point cloud branch takes geometric mapping images GMI_{train} as input, also using ResNet18 architecture to extract spatial structural information from point clouds.

It's worth noting that attention mechanisms have been increasingly applied in deep learning in recent years [21]. Considering that most regions of GMI_{train} contain less effective information due to point cloud sparsity, to enhance the contribution of dense regions to final calibration prediction, this paper introduces a spatial attention module after point cloud branch feature extraction. This module calculates global statistical information of feature maps and adopts a max-min normalization strategy to generate spatial attention maps, effectively enhancing the feature expression capability of key regions.

For input feature map $x \in R^{C\times H\times W}$, the spatial attention weights are calculated by the following formula:

$$AttentionMap(i,j) = \frac{\frac{1}{C}\sum_{c=1}^{C}|x_{c,i,j}| - \min_{i,j}(\frac{1}{C}\sum_{c=1}^{C}|x_{c,i,j}|)}{\max_{i,j}\left(\frac{1}{C}\sum_{c=1}^{C}|x_{c,i,j}|\right) - \min_{i,j}\left(\frac{1}{C}\sum_{c=1}^{C}|x_{c,i,j}|\right)} \tag{14}$$

where $x_{c,i,j}$ represents the value of the input feature map at channel c position (i,j); C is the total number of channels of the input feature map. Subsequently, this attention map is used to weight the original feature map:

$$x'_{c,i,j} = x_{c,i,j} \cdot AttentionMap(i,j) \tag{15}$$

where $x'_{c,i,j}$ is the feature map after attention weighting.

Through this attention mechanism, the feature expression capability of point cloud dense regions can be effectively enhanced, interference from invalid regions can be weakened, thereby improving the contribution of point cloud features in subsequent feature fusion processes.

Feature Fusion and Processing: To effectively fuse cross-modal feature information from images and point clouds, this paper concatenates features extracted from two branches in the channel dimension and introduces additional convolutional modules for feature refinement, gradually achieving fusion, dimensionality reduction, and optimization of multi-modal information.

Fully Connected Regression Layer: The fused features are input to the regression module after flattening to predict rotation $r = [\theta, \varphi, \alpha]$ and translation parameters $t = [x, y, z]$。

Figure 5 intuitively shows the deep learning network structure proposed in this paper.

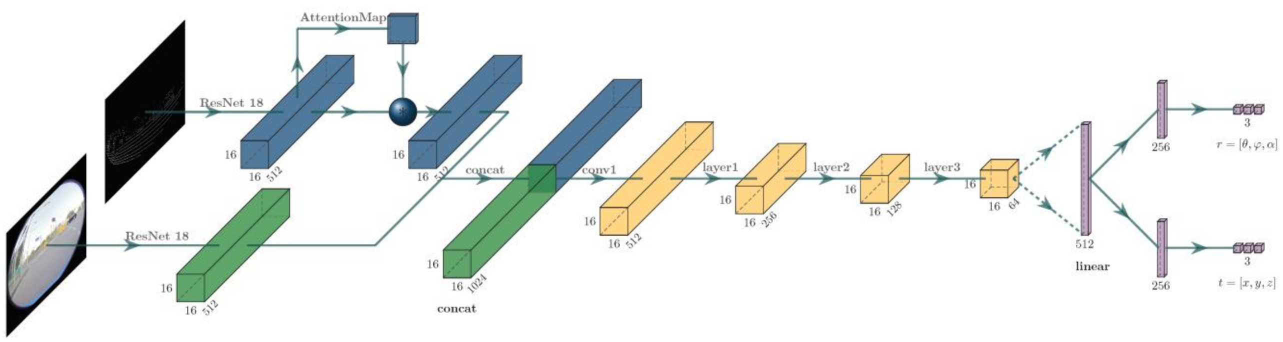

Fig.5 Deep Learning Network Architecture

2) Loss Function:

To improve the accuracy of extrinsic estimation, the loss function designed in this paper is based on point cloud alignment error, used to measure the difference between point clouds transformed using predicted extrinsics and ground truth extrinsics. This loss is essentially equivalent to the square of root mean square error, used to measure the overall spatial deviation of point clouds:

$$loss_{lidar} = \frac{1}{N} ||M_{pre}^{-1} P_{train}^{(h)} - M_{label}^{-1} P_{train}^{(h)}||_F^2 \tag{16}$$

where $||\cdot||_F^2$ represents the square of Frobenius norm, i.e., the sum of squares of all elements in the matrix; N is the total number of points in the point cloud; M_{pre} represents the extrinsic transformation matrix predicted by the deep learning model, converted from six parameters (rotation and translation) output by the model; M_{label} is the ground truth extrinsic matrix artificially set during training sample generation, used to supervise model learning.

3) Geometric Optimization:

Although deep learning-based extrinsic estimation methods perform well in LiDAR and fisheye camera calibration tasks, due to the relatively sparse LiDAR point clouds used, the model may still experience accuracy degradation in certain specific scenarios. To further improve the accuracy and practicality of extrinsic estimation, this paper introduces a geometric optimization strategy to enhance the matching quality between point clouds and images.

The specific optimization process is as follows: Define the image-point cloud pair set for the geometric optimization stage as: $\Theta_{post} = \{(I_{post_i}, P_{post_i})\}$, where I_{post_i} represents the fisheye image of the i-th frame, P_{post_i} represents the original LiDAR point cloud of the i-th frame transformed using the initial extrinsic matrix M_{init}.

To ensure the effectiveness of geometric optimization data, the following conditions must be satisfied: background information is sufficiently exposed within a certain time window, and foreground targets are continuously visible and unoccluded in the sensor field of view. This condition ensures that foreground targets are accurately extracted from images, providing reliable basis for point cloud matching. In this paper, following personnel as foreground targets satisfy this condition.

First, foreground targets need to be extracted from I_{post_i}. This study adopts a background/foreground segmentation algorithm based on Gaussian mixture models [22], and performs undistortion processing on segmentation results to obtain more accurate foreground regions. The foreground extraction process can be expressed as:

$$F_{post_i} = G(I_{post_i}) \tag{17}$$

where $G(\cdot)$ is the foreground extraction algorithm; F_{post_i} represents the foreground person region extracted from the i-th frame image I_{post_i}.

Subsequently, to obtain foreground targets from P_{post_i}, the following steps are taken:

$$M_{pre_i} = \mathcal{F}_{net}(\Theta_{post}) \tag{18}$$

$$P_{pre_i}^{(h)} = M_{pre_i}^{-1} P_{post_i}^{(h)} \tag{19}$$

$$P_{fg_i} = Cluster(\{P_{pre_i} | p \in F_{post_i}\}) \tag{20}$$

$$P_{fg_i}^{*} = M_{pre_i} P_{fg_i} \tag{21}$$

where $\mathcal{F}_{net}(\cdot)$ is the trained deep learning network function; M_{pre_i} is the extrinsic matrix obtained by inputting Θ_{post} into the deep learning model; $Cluster(\cdot)$ represents clustering algorithm; $P_{fg_i}^{*}$ is the final foreground point cloud containing only points corresponding to foreground targets.

As shown in Figure 6, after the above processing steps, the original image-point cloud pair $\Theta_{post} = \{(I_{post_i}, P_{post_i})\}$ can be optimized to data pair $\Theta^{*}_{post} = \{(F_{post_i}, P^{*}_{fg_i})\}$ for subsequent optimization stages.

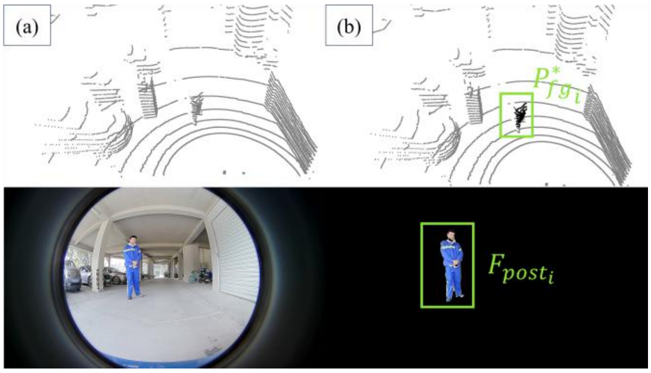

Fig.6 Illustration of Data Extraction for Geometric Optimization

urthermore, based on the extrinsic parameter representation form mentioned above, namely $\theta, \varphi, \alpha, x, y, z$, the following optimization objective function can be constructed:

$$\max_{\theta,\varphi,\alpha,x,y,z} \sum_{i} \sum_{p \in P^{*}_{fg_i}} \mathbf{1}(\Pi(K \cdot p) \epsilon F_{post_i}) \tag{22}$$

where $\Pi(K \cdot p)$ represents projecting point p to the image plane through camera intrinsics; $\mathbf{1}(\cdot)$ is an indicator function that takes value 1 when the projected point falls within F_{post_i}, otherwise 0.

2 Results Analysis

2.1 Qualitative Evaluation

To verify the actual effectiveness of the proposed deep learning model in extrinsic estimation tasks, Figure 7 shows visualization results of multiple point clouds projected onto images under different extrinsic conditions. From the figure, it can be observed that under initial extrinsic conditions, there are obvious spatial deviations between point cloud projections and images; after correction with extrinsics predicted by the model, the spatial alignment between point clouds and images is significantly improved, indicating that the model has effectively learned the geometric mapping relationship between LiDAR and fisheye cameras.

This paper further introduces geometric optimization strategies to improve the accuracy of final extrinsic estimation. Figure 8 shows comparison results of point cloud projections under three conditions: initial extrinsics, deep learning predicted extrinsics, and geometrically optimized extrinsics. From the figure, it can be clearly seen that point clouds after geometric optimization have the best alignment with images, further verifying the effectiveness of the geometric optimization strategy proposed in this paper from a visual perspective.

2.2 Quantitative Evaluation

To further evaluate the quantitative performance of the model in extrinsic estimation tasks, this paper uses randomly generated ground truth extrinsic matrices MlabelM_{label} Mlabel to calculate point cloud alignment loss and conducts comparative analysis with deep learning model prediction results. The point cloud alignment loss before model inference is defined following equation (16), namely the square form of root mean square error:

$$loss'_{lidar} = \frac{1}{N}||P_{train}^{(h)} - M_{label}^{-1} P_{train}^{(h)}||_F^2 \tag{23}$$

where N represents the total number of three-dimensional points in the point cloud.

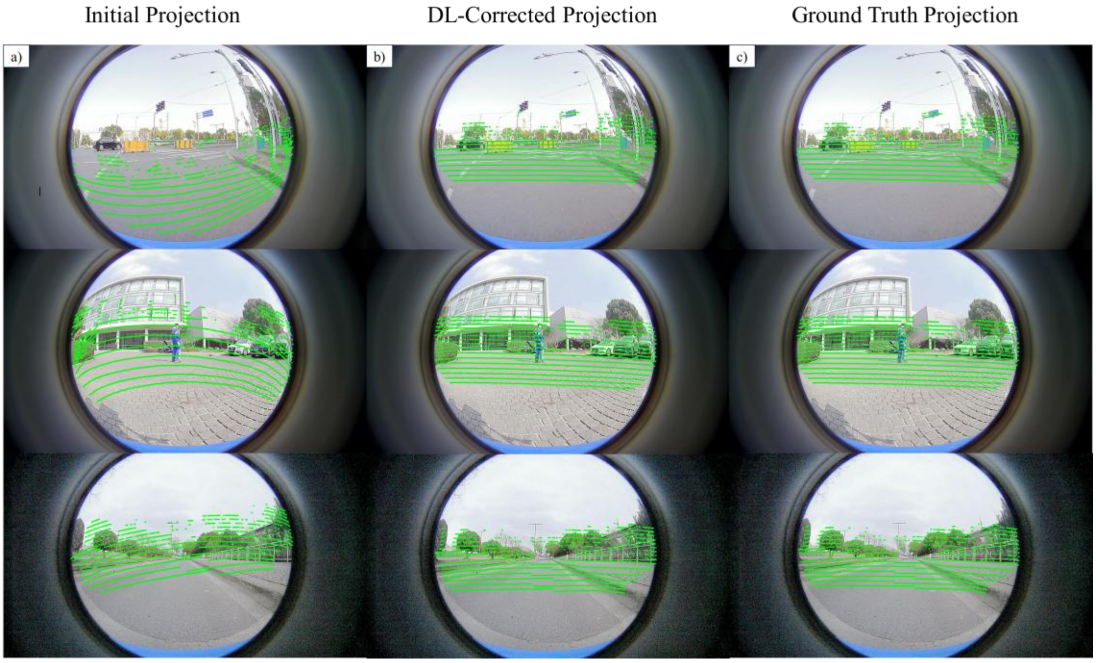

Fig.7 Visualization Comparison of Point Cloud Projection Before and After Deep Learning Correction

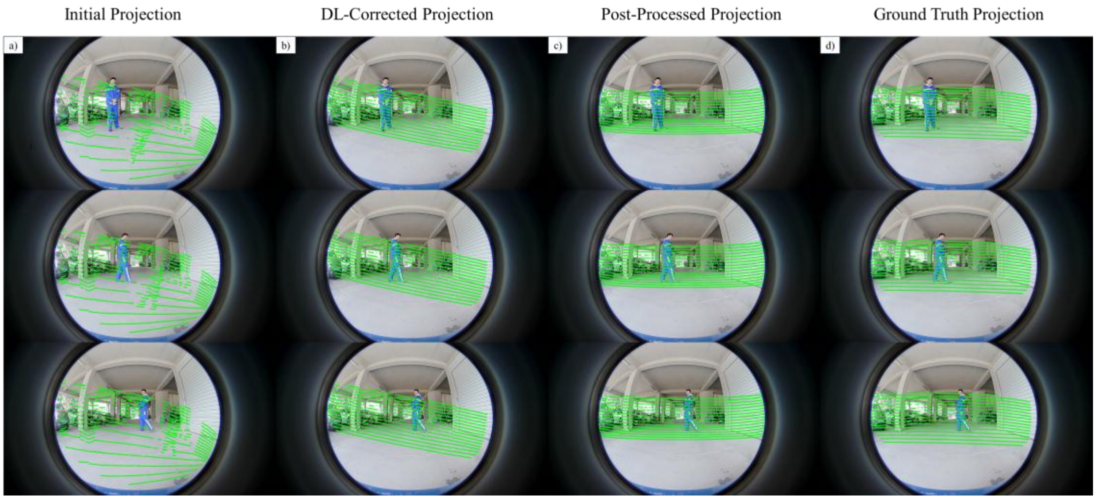

Fig.8 Visualization Comparison of Point Cloud Projection Before and After Geometric Optimization

Figure 9 shows the changes in point cloud alignment loss before and after deep learning correction for a total of 18,335 data groups. The results show that the average error significantly decreased from 6.517839 to 0.548611, with an error reduction of approximately 91.6%, indicating that the model has high accuracy in learning spatial geometric relationships between point clouds and images.

Furthermore, Figure 10, based on the image-point cloud pair set $\Theta_{post} = \{(I_{post_i}, P_{post_i})\}$, used in the geometric optimization stage, statistics the point cloud alignment errors in three stages: initial state, after deep learning prediction, and after geometric optimization. The results show: the initial average error was 2.733867, which decreased to 0.584133 after model prediction, with an error reduction of 78.4%; based on this, geometric optimization further reduced the error to 0.278217, an additional decrease of 52.4%. This result verifies the practical value and effectiveness of the geometric optimization strategy in improving calibration accuracy.

In summary, the extrinsic calibration method proposed in this paper not only achieves good alignment between point clouds and images at the visual level, but also demonstrates excellent accuracy and robustness in quantitative evaluation, fully proving its application feasibility and practical value in real scenarios such as automatic following vehicles.

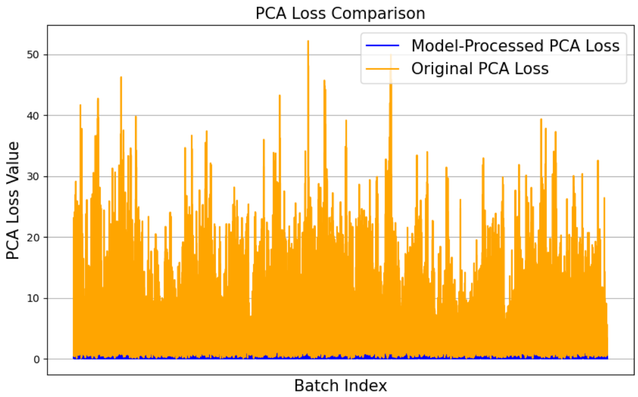

Fig.9 Comparison of Point Cloud Alignment Loss Before and After Deep Learning
(PCAL: Abbreviation for Point Cloud Alignment Loss)

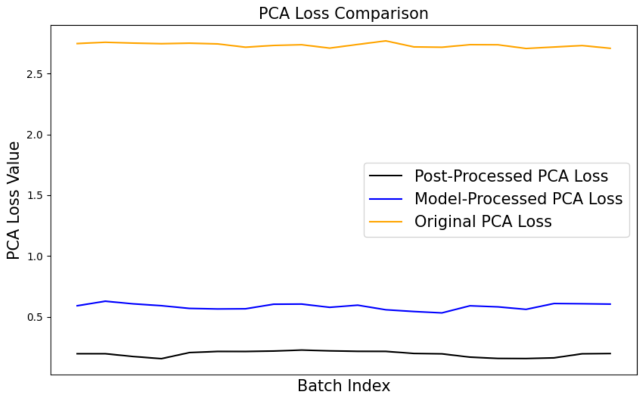

Fig.10 Comparison of Point Cloud Alignment Loss Before and After Deep Learning and Geometric Optimization

3 Conclusion

This paper proposes an extrinsic calibration method for LiDAR and fisheye cameras combining deep learning and geometric optimization, aimed at significantly improving alignment accuracy between point clouds and images. The method constructs a regression model based on deep neural networks that can directly predict extrinsic matrices from image and point cloud data, and is trained through carefully designed point cloud alignment loss functions, effectively reducing projection errors of point clouds in fisheye images. In the geometric optimization stage, a foreground target extraction strategy is introduced, optimizing extrinsic parameters by filtering point cloud data corresponding to target regions, further improving calibration accuracy.

Experimental results show that this method demonstrates good generalization ability and robustness under various environmental conditions, with broad application potential in multi-sensor fusion scenarios of autonomous driving systems.

Future research will focus on improving the stability of this method under extreme environments (such as low illumination, dynamic occlusion, etc.), and further exploring data fusion mechanisms with other sensors such as IMU to improve extrinsic estimation accuracy. Additionally, combining lightweight neural network design can reduce model computational resource requirements and enhance deployment efficiency on embedded platforms and real-time systems.

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